

SHUAIKE SHEN

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EDUCATION

Carnegie Mellon University

Ph.D in Computational Biology

Pittsburgh, PA, U.S.

Incoming

Zhejiang University

Undergraduate Student in Computer Science and Technology

Hangzhou, Zhejiang, China

Sept. 2021 - Present

- Mixed Class, [Chu Kochen Honors College](#)
- GPA: 3.83/4.00

RESEARCH INTEREST

I am deeply interested in research areas related to deep generative models, bioinformatics, and AI for science (AI4Science). My previous research experience includes protein design, motif scaffolding, and the prediction of crystal structures and properties. Currently, I am focusing on two main topics: The first involves latent flow matching for protein design, which offers both greater efficiency and scalability. The second explores the application of reinforcement learning to flow matching models, with the goal of aligning human reward signals. This approach can be leveraged to design proteins with diverse substructures and other biomolecules exhibiting high affinity.

RESEARCH EXPERIENCE

Research Intern

Advisor: [Prof. Ge Liu](#) and [Prof. Nicholas Ching Hai Wu](#)

University of Illinois Urbana-Champaign

June. 2024 - Present

- Utilized latent flow matching for joint protein structure and sequence design. Most previous models treat structure and sequence design as separate stages, leading to inconsistencies between the two. The latent codesign paradigm addresses this issue by learning a joint latent space that captures the information from both modalities. Additionally, by applying the latent space to flow matching, we can scale up the model, achieving improved performance.
- Applied reinforcement learning to flow matching models to align with human-defined rewards. By introducing a regularization method, this RL framework addresses the issue of model collapse, a common challenge in RL fine-tuning. I implemented this framework in protein design tasks, with the goal of designing novel and diverse protein structures featuring various substructures. Additionally, the framework can be applied to design stable biomolecules with high affinity.

Undergraduate Research Assistant

Advisor: [Prof. Chunhua Shen](#) and [Prof. Hao Chen](#)

Zhejiang University

Oct. 2022 - May. 2024

- Contributed to *VFN*, a novel SE(3) invariant graph encoder, addressing the atom representation bottleneck observed in traditional IPA encoders. In both Inverse Folding and Protein Design tasks, this innovative encoder consistently outperformed its predecessor.
- Purpose *FADiff*, a protein multi-motif scaffolding endeavor. *FADiff* treats multiple motifs as rigid bodies capable of rotational and translational movements in space. The model developed for this project successfully accomplishes the task of multi-motif scaffolding and has exhibited excellent performance across various evaluation metrics. And *FADiff* has extraordinary generalization ability: model trained on two motifs can scaffold up to five motifs during inference.
- Purpose a denoising pre-training frameworks (*DPF*) on crystals to boost the model's ability to extract the general relationship between crystal structure and properties, which can take advantage of unlabeled crystal structure data instead of only a small amount of labeled structure.
- Proposed *RiFold* to improve the existing RNA Inverse Folding model. RNA exhibits typical complementary base pairing, so one-shot decoding can lead to incorrect base pairing. To address this, I proposed using autoregressive decoding to ensure correct base pairing. *RiFold* outperforms all baseline models and achieves state-of-the-art results across multiple benchmarks.
- Proposed *CEBind*, an unsupervised binding energy prediction model based on the law of conservation of energy, which states that the total energy of an isolated protein complex system is constant. This model offers a fast and efficient method for predicting binding energy in complexes, with potential applications in complex design.

PUBLICATIONS

* denotes equal contribution and † denotes corresponding author.

- From Sentences to Sequences: Rethinking Languages in Biological System** July. 2025
- Ke Liu*, **Shuaike Shen***, Hao Chen†
 - Under review
 - Arxiv, [\[Link\]](#)
- Online Reward-Weighted Fine-Tuning of Flow Matching with Wasserstein Regularization** Jan. 2025
- Jiajun Fan, **Shuaike Shen**, Chaoran Cheng, Yuxin Chen, Chumeng Liang, Ge Liu†
 - **ICLR 2025**, [\[Link\]](#)
- A Denoising Pre-training Framework for Accelerating Novel Material Discovery** Dec. 2024
- **Shuaike Shen**, Ke Liu, Hao Chen†
 - **AAAI 2025 AISI**
- Boost Your Crystal Model with Denoising Pre-training** June. 2024
- **Shuaike Shen***, Ke Liu*, Muzhi Zhu, Hao Chen†
 - **ICML 2024 AI for Science Workshop**, [\[Link\]](#)
 - Project Page, [\[Link\]](#)
- FADiff: Floating Anchor Diffusion Model for Multi-motif Scaffolding** May. 2024
- Ke Liu*, **Shuaike Shen***, Weian Mao*, Xiaoran Jiao, Zheng Sun, Hao Chen†, Chunhua Shen†
 - **ICML 2024**, [\[Link\]](#)
 - Project Page, [\[Link\]](#)
- De novo protein design using geometric vector field networks** Oct. 2023
- Weian Mao*, Muzhi Zhu*, Zheng Sun*, **Shuaike Shen**, Lin Yuanbo Wu, Hao Chen†, Chunhua Shen†
 - **ICLR 2024 spotlight**, [\[Link\]](#)
 - Project Page, [\[Link\]](#)
- Training Text-to-Image Flow Models on Self-Generated Data** Nov. 2024
- Jiajun Fan, **Shuaike Shen**, Chaoran Cheng, Chumeng Liang, Ge Liu†
 - Under review
- Physics-Informed Neural Networks for Unsupervised Binding Energy Prediction** Aug. 2024
- Ke Liu*, **Shuaike Shen***, Hao Chen†
 - Under review
 - Manuscript, [\[Link\]](#)

SELECTED PROJECTS

- VITON: High resolution virtual try on based on generative models 2023
Scientific research projects of Chu Kochen Honors College advised by [Prof. Hao Chen](#)
- MiniSQL: A database system implemented in C++ 2023
The course project of Database systems and concepts
- Parallel optimization of big data pivots selection based on vectorization and OpenMP 2022
High Performance Computing 101 summer course | ACM-China IPCC 2022

SKILLS

- Programming Skills:** Python, C/C++, PyTorch, SQL
- Language Skills:** Mandarin (Native), English (TOEFL 104)